One Policy per Sequence, Selected Without Ground Truth

Extended Abstract

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Abstract: We enter all three tiers of Track 2 with one FlashCHORD policy per sequence on a Unitree G1 with Dex3 hands. The tiers share a ground truth and differ only in the *input* a submission may condition on, which makes the permitted input the design variable rather than the reward. Everything per-clip that we ship therefore reads the input alone: a registered selection rule reproducible without ground truth, and, on Tier 1, a warm-start fine-tune re-fitting a converged actor to the cleaner reference that tier permits — the one method change that paid, at $0.57 per clip, where a cold swap of the same reference *loses* 0.18 AUC.

1. Introduction

Track 2 asks for physically realisable object trajectories for 30 manipulation sequences, produced by a policy rollout rather than by reconstruction, at three tiers differing in the reference a submission may condition on: a clean multi-view trajectory (Tier 1), a noisy released retarget (Tier 2), and whatever the entrant reconstructs from video (Tier 3).

Our first result was structural and we found it by accident: our Tier 1 and Tier 2 public scores came back *identical*. The uploaded files are byte-identical, and a Kabsch fit shows every shipped rollout's reference to be an exact (0.000 cm) rigid transform of the released Tier 2 trajectory. All three tiers are scored against the same ground truth; only the permitted input differs.

That set the strategy: the interesting constraint is not the reward but *what a submission may read*. Relative to physics-based human–object imitation [4, 6], which assumes a clean motion-capture reference, the difficulty here is that the reference is itself an estimate.

Contributions:

- a per-episode selection rule reading only the *input* reference: legal in every tier, reproducible without ground truth;
- the measurement that a converged policy cannot be handed a cleaner reference — a cold swap costs 0.18 AUC — while a 50M-step warm-start fine-tune to it gains 0.08 for $0.57.

2. Approach

Pipeline. SOMA-X parameters are retargeted to the G1 with Dex3 hands on CPU (30/30 clips at 0.0 mm residual). FlashCHORD SAC then trains one policy per sequence for 349,999,104 steps at `world_count=1`, `scene.support` set per clip from the shipped primitives, followed by the challenge evaluator, selection and the kit's packer.

Recipe changes. Four ship: an object-*orientation* reward term (weight 1.0, std 0.3), the configuration having a position weight and no orientation counterpart; a start-curriculum patch; object-position std tightened 0.2 → 0.08 m; and finger targets on bridge-smoothed references. Each *refuses to apply* to an unrecognised upstream, so a half-patched run cannot reach the GPU.

Selection, registered before the run. The provenance schema permits a different `checkpoint_sha256` per episode, so we pick, per clip, the latest checkpoint whose achieved object never leaves a 10 cm band around the *input* reference. It cost us real score: 13 rollouts kit-score higher than the ones that ship, and we shipped the rule's picks anyway.

Tier 3. We run the challenge's own modules in order — monocular geometry [3], open-set detection [1], segmentation [2] and 6D tracking [5] — with a prompt converter written between the last two. Placement in the released world is one rigid transform taken from frame 0, as the FAQ permits, applied unchanged thereafter: no scale fitted, no per-frame correction, 0.0000 cm frame-0 residual on all 30 clips.

Warm start. A new configuration key loads a shipped actor's parameters and batch statistics after learner construction, sets the step counter so the curriculum resumes in its final stage, and runs to the

Entry	*AUC*	*SP-SR*	*MP-SR*	*MPPE*
Tier 1 (ours)	**0.548**	0.700	0.500	**12.06**
CHORD base.	0.510	—	—	—
Tier 2 (ours)	**0.465**	0.667	0.467	**14.13**
CHORD base.	0.374	0.533	0.067	20.50
Tier 3 (ours)	0.268	0.200	0.033	47.77

Table 1: All 30 clips, one seed, official kit scorer; MPPE in cm, lower better.

normal horizon — 50M steps, a seventh of a full run, 35 minutes on an A6000. The critic starts cold, so the actor's learning rate is scaled $\times 0.1$ for 2,000 updates.

Scoring. We never reimplemented the metrics: our scorer hash-checks the kit's metric files against the shipped manifest and calls them, so across nineteen uploads the largest local-to-public AUC disagreement was 1.1×10^{-5}.

3. Experiments and Results

The largest single effect was a bug. Our first full-length baseline scored 0.043 mean AUC because every run script passed `scene.support=false`: without the table the object falls at reset and the episode ends on frame 0.

A cleaner reference is not an upgrade; re-fitting to it is. Tier 1 permits the clean multi-view trajectory, so we re-evaluated all 30 shipped actors on it directly. Mean AUC *fell* from 0.465 to 0.289 and SP-SR from 0.667 to 0.138, with only 2 of 30 clips improving; frame checks were below 1.7×10^{-5} cm, so this is not a conversion artefact: each actor had learned one trajectory, and substituting another is distribution shift. Warm-start fine-tuning on that same reference, under a no-regression adoption rule, instead took Tier 1 to 0.548 AUC and 12.06 cm. The gain is re-fitting, not extra training — a second generation on the *same* reference adopted 0 of 3 and lost a clip outright ($0.647 \rightarrow 0.296$) — and is predicted by a clip's shipped score: 11 of 19 clips above 0.35 adopted, mean gain +0.169; 0 of 4 below 0.35 did. On Tier 3, where selection may read only our own reconstruction, 5 of 10 adopted, moving the entry from 0.232 to 0.251 and lifting both success rates off zero.

Selection is worth less than it looks. Across all 30 clips the registered rule is worth +0.0010 AUC over the final checkpoint, and a truth-reading oracle — the ceiling on *any* selection rule — reaches only 0.420. It earns its place by being reproducible without ground truth, not by the points it adds. The standard deviation of AUC over a clip's last five checkpoints averages 0.081, a noise floor larger than most effects we measured.

Where the gap is. Measured on trajectories rather than on AUC, which mixes position and orientation so two defects can cancel: the noisy Tier 2 reference sits 1.94 cm from truth while our policies sit 13.84 cm, and *19 of 30 clips end up worse than the reference they were given.* Yet only 9 ever leave a 10 cm band around it — nine drop catastrophically and the rest track a good reference a few centimetres loosely, so precision across many clips, not robustness on a few, is where the deficit lives. Tier 3 is the opposite: eleven clips have *negative* headroom, the policy already beating the reconstruction it was handed.

4. Conclusion

A benchmark that varies only the permitted input rewards treating that input as the design variable, and the two results above transfer beyond this challenge. The residual failure does not: it is mid-episode release of a well-tracked object, and one second before each loss nothing observable differs from a healthy frame. The cause appears to be that the environment computes a per-link contact state every step, pays a reward term carrying the largest positive weight in the objective from it, then discards it before building the observation — so the actor is paid for contact it cannot sense, and the finger action through those frames is open-loop. That is the correction the contact-conditioned literature reports [4, 6], and the first thing we would run next.

At scale. Spend was $144 for 30 policies across three tiers — $3.00 per full-length cell, $0.57 per warm-start cell — so a new sequence costs little and a new *reference* for an existing one almost nothing (*economic viability*). Selection and adoption read only the input, so rollouts are judged with no annotation and no ground truth: the loop closes without a human in it (*data flywheel*). One CLI, patches that refuse unknown upstreams and a scorer calling the kit's own metric code under a hash check keep it auditable (*complexity reduction*); the input-only rule and warm-start reference transfer are new here (*innovation*).

Limitations. One seed, and no ablation isolates the four recipe changes: the shipping run was chosen as such, knowingly. Object inertia is a simulator artefact we left alone.

References

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